TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Standardization of Sodium Hydroxide with a Standard Oxalic Acid Solution

Why NaOH cannot be weighed into accuracy, the titration against the primary standard oxalic acid, the indicator choice, and the full calculation.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 13 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to determine the exact concentration of an approximately prepared NaOH solution. NaOH is a **secondary standard** (Experiment 11): it absorbs water and carbon dioxide from the air, so no weighing of it is trustworthy. It is therefore made approximately (say 0.1 M) and standardised against a **primary standard acid** — oxalic acid, which can be weighed exactly. The reaction neutralises both of oxalic acid's protons:

$$H_2C_2O_4 + 2\ NaOH \rightarrow Na_2C_2O_4 + 2\ H_2O \quad \text{(1 mol acid : 2 mol alkali)}$$

Indicator: phenolphthalein. The titration is weak acid versus strong base, so the equivalence point is alkaline (the oxalate ion hydrolyses — Chemistry-I Topic 3), inside phenolphthalein's range (colourless → pale pink) and far from methyl orange's. Choosing the indicator is applying Topic 3, not decorating the flask.

2. Apparatus and solutions

- Standard oxalic acid solution (0.05 M prepared from the dihydrate as in Experiment 11 — 1.575 g per 250 mL for 0.05 M… note: 0.05 M × 0.250 L × 126 g mol^{-1} = 1.575 g exactly), pipette 25 mL, burette, conical flask, phenolphthalein.
- The NaOH solution under standardisation fills the burette (it is the unknown being measured; the known acid is pipetted into the flask).

3. Procedure

1. Rinse the burette with the NaOH solution and fill it; remove the funnel; read the initial level (meniscus at eye level, no air bubble in the jet).

2. Pipette 25.0 mL of the standard oxalic acid into a clean conical flask; add 2–3 drops of phenolphthalein.
3. Titrate: add NaOH with swirling, slowing to drops near the end, until one drop turns the solution **permanently pale pink** (30 seconds' persistence). Record the final reading.
4. Repeat until two titres agree within 0.1 mL; average the concordant titres.

4. Worked calculation (model data)

STANDARDISATION RESULT

Standard oxalic acid: 0.0500 M. Volume pipetted: 25.0 mL. Concordant titre of NaOH: 24.6 mL.
Moles of oxalic acid = $0.0500 \times 0.0250 = 1.25 \times 10^{-3}$ mol.
Moles of NaOH = $2 \times 1.25 \times 10^{-3} = 2.50 \times 10^{-3}$ mol (the 1 : 2 ratio is the step students lose marks on).
Molarity of NaOH = $2.50 \times 10^{-3} / 0.0246$ = **0.1016 M**.
In normality terms: 0.05 M oxalic acid = 0.10 N; N(NaOH) = 0.10 × 25.0/24.6 = 0.1016 N — the same answer by the equivalents road, as it must be (NaOH is monobasic, so M = N).

5. Observation table (as recorded)

TRIAL	INITIAL READING (ML)	FINAL READING (ML)	TITRE (ML)
1 (rough)	0.0	25.1	25.1
2	0.0	24.6	24.6
3	0.0	24.6	24.6

The rough titre locates the endpoint; the concordant pair does the science. Report the standardised value with its date on the bottle — a standardisation ages (NaOH keeps absorbing CO_2) and is repeated when results matter.

6. Precautions

- The flask is rinsed with distilled water only — never with the acid (extra acid would consume extra alkali invisibly). The pipette and burette are rinsed with their own solutions.
- Distilled water used in this titration should be CO_2-free (freshly boiled and cooled) for the finest work: dissolved CO_2 is an acid that phenolphthalein counts.

- Read the burette without parallax; remove the funnel before titrating (it drips); swirl continuously near the endpoint and wash down the flask's walls.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is oxalic acid, and not NaOH itself, the primary standard in this experiment?
2. Why phenolphthalein and not methyl orange? Answer through the equivalence-point pH.
3. 25.0 mL of 0.0500 M oxalic acid needs 25.8 mL of NaOH. Find the NaOH molarity.
4. A student rinses the conical flask with the oxalic acid solution before pipetting into it. What error follows, and in which direction?
5. Why are two concordant titres required rather than one careful one?
6. State the mole ratio of the reaction and the single most common calculation error it causes.

Answer key

1. Oxalic acid dihydrate meets the primary-standard tests (pure, stable in air, definite composition, weighable exactly). NaOH fails them — it absorbs moisture and CO_2, so its weighed mass is not a known amount of alkali; only titration against a primary standard can certify it.
2. The weak acid–strong base equivalence point is alkaline (oxalate hydrolyses), around pH 8–9 — within phenolphthalein's change range (8.2–10). Methyl orange changes at pH 3–4.4, far before equivalence: it would signal an endpoint with acid still un-neutralised.
3. Moles acid = $0.05 \times 0.025 = 1.25 \times 10^{-3}$; moles NaOH = 2.50×10^{-3}; M = $2.50 \times 10^{-3}/0.0258 = 0.0969$ M.
4. The rinse adds unmeasured oxalic acid to the flask, so more NaOH is consumed; the calculation credits that alkali to the pipetted 25 mL alone and reports the NaOH stronger than it is (molarity overestimated).
5. A single titre can hide a blunder (overshoot, misread, bubble). Concordance of repeats is the experiment's evidence that the number is a property of the solutions, not of one performance.
6. 1 mol oxalic acid : 2 mol NaOH (both protons are neutralised). The common error is to use 1 : 1, halving the computed NaOH concentration.